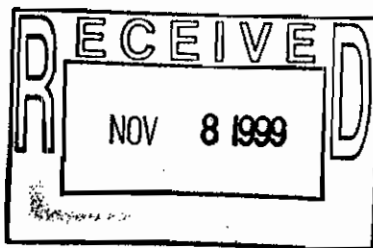


**ATTACHMENT 11 – RESPONSE LETTER FROM MODERN CONTINENTAL
COMPANY TO BECHTEL / PARSONS BRINCKERHOFF DATED
NOVEMBER 8, 1999**
(7 pages)

November 8, 1999

Central Artery/Tunnel Project
Bechtel/Parsons Brinkerhoff
185 Kneeland Street
Boston, MA 02111



Attention: Mr. R. E. Steffy
Authorized Representative

Subject: Central Artery (I-93) / Tunnel (I-90) Project
Contract No. 99119-C09B2
I-90 Seaport Access Tunnel Finishes
Adhesive Anchor Movement

Reference: B/PB Letter No. C09B2-0350, dated October 12, 1999

Dear Mr. Steffy:

With regard to the adhesive anchor movement observed in Ramp HOV, Modern Continental Construction offers the following in response.

In response to the observed anchor movement, and pursuant to discussions with the Engineer regarding this matter, Modern Continental Construction invited a technical representative from Powers-Rawl Fasteners to the site in order to assess the current condition of the anchors. This site visit was conducted on October 21, 1999. A summary of this site visit and recommendations for corrective action has been attached for your review.

In the above referenced correspondence, the Engineer cites improper ceiling module installation and anchor hole preparation procedures as potential causes for the anchor movement. Modern Continental and Powers-Rawl do not agree with these statements. If the anchors were improperly installed (i.e. drill hole preparation, epoxy installation, etc.), it is likely that the anchors would have failed the proof test. As all of the anchors passed proof testing prior to module installation, it is not likely that anchor installation procedures were the cause of the anchor movement. Further, the ceiling modules were erected in accordance with the approved erection procedure, and in the presence of the Engineer. The assertion that the lifting devices were removed too quickly is unfounded.

Modern Continental Construction is currently conducting a review of the engineering associated with the adhesive anchors. Given the parameters associated with the ceiling design (approximate module weight, number of anchors supporting the module, and tensile capacity of the anchor) and the applied factors of safety, it is improbable that the anchors were overstressed as a result of erection loads or the erection procedures. While the measured torque on some anchors exceeded recommended values, the manufacturer states that these anchors are functional, and should be subjected to a proof-load test for verification.

Letter No. C09B2-0139

Page 2 of 2

In summary, Modern Continental Construction confirms that all elements of the ceiling have been installed in accordance with the Contract Documents. Per the recommendations of Powers-Rawl, Modern Continental Construction will remove and replace the anchors which exhibited movement, and will proof test the anchors subjected to torque values higher than the manufacturer's recommended values in order to verify the anchor capacity. Modern Continental is currently evaluating available personnel and resources necessary to undertake this effort, and will inform the Project prior to the start of this operation.

However, in that the anchors have already passed the proof test and the root cause of the anchor failure has yet to be determined, Modern Continental considers this additional testing and support of this operation to be a change to the Contract. In accordance with Division I, Section 4.03, Modern Continental Construction reserves our rights to claim for additional costs associated with this change.

Sincerely,

Modern Continental Construction Company Inc.



Robert Coutts
Sr. Project Manager

RC/NJ

cc: M. Charney, B/PB
C. Madden
J. Pastore

Powers FASTENERS

Friday, October 29, 1999

The Modern Continental Construction Co., Inc.
6 Necco Court
Boston MA 02210

Attn: Mr. Neil Joyce
Ph: (617) 695-2277
Fax: (617) 695-2272

Powers Fasteners, Inc.
2 Powers Square
New Rochelle, NY
USA 10801

Telephone: 1-(914) 235-6300
Fax: 1-(914) 676-6483

Web Site: www.powers.com
E-Mail: williamson@powers.com

Re: CO9B2 I-90 Seaport Access

Dear Mr. Joyce:

Thank you for allowing Powers' Regional Representative, Mr. Jack Armour and Powers' Field Engineering Manager, Mr. Stephen Rice access to the above mentioned job site to inspect the six anchors that showed movement in the HOV ceiling mock up.

The following job site information was provided to Powers Fasteners by others:

- a) over (1000) 5/8" diameter Power-Fast adhesive anchors were installed in diamond cored anchor holes to a minimum embedment depth of 5"
- b) the anchor rod material was Type 316 Stainless Steel threaded rod
- c) anchor seal plugs were implemented as part of the adhesive anchoring system per installation recommendations due to overhead anchorage locations
- d) base material was reported to be solid, normal weight concrete having a minimum compressive strength of 4,000 psi
- e) adhesive anchors were installed in July of 1999.
- f) within two or three days after installation, all of the adhesive anchors were proof load tested to a load of 3,250 lbs in tension in accordance with procedures approved by the Project Engineer that followed the guidelines of ASTM E-488.

Conam Inspection, Inc.'s Field Test Reports dated July 30 through August 16, 1999 had been reviewed prior to Mr. Rice's arrival to the site. Mr. Rice was told while on site that anchors reported as failures in Conam Inspection's Reports were replaced, retested and approved prior to installation of the HOV ceiling mock up.

In Mr. Robert Coutts of Modern Continental's letter of October 7, 1999 addressed to Mr. R.E. Steffy representing the Central Artery Tunnel Project, it was reported that approximately one month after the HOV ceiling mock up installation, initial movement of about 1/16" was observed on the six anchors in question. This condition was monitored for the next six weeks whereas the worst case of anchor movement was reported to be approximately 9/16". This information also corresponds to the condition reported in your

letter of October 19, 1999. No additional anchor movement since October 19th has been reported.

Upon visual inspection of the anchors in question, Mr. Rice inquired as to details of the anchor installation as well as the installation procedure of the ceiling hanger plates. The main question raised, "Why would an anchor that was successfully proof loaded and believed to be correctly installed, move after the ceiling hanger plates were installed?"

Mr. Rice reported that he proceeded to check the applied torque on the nuts of the anchors with a torque wrench. Mr. Rice checked both the anchors that showed movement as well as anchors that had not moved and got torque readings of 120 foot-pounds. This torque value is greater than the published maximum torque range of a 5/8" diameter Power-Fast adhesive anchor (50-90 foot-pounds). The maximum torque ranges published for Powers' injection adhesive systems are based on testing performed in normal weight concrete at nine diameters of embedment depth.

Answers to your general questions of October 19th are as follows:

- Please refer to the attached excerpt from Powers' *Fastening Systems Design Manual, Second Edition*, Section 22.4.1 entitled "Displacement". Curve # 1 in the illustration shows the typical performance of an adhesive type anchor. If an appropriate factor of safety is applied to the ultimate load capacity of the adhesive, then displacement should not occur provided that proper installation procedures were employed.
- Any movement in the anchor may be an indication that the allowable load capacity of that particular installation has potentially been exceeded. As shown in Curve # 1, when displacement becomes more significant, the anchor approaches its ultimate failure load capacity.
- If a problem during installation occurred, it is possible that an anchor could pass a proof load test provided a large enough safety factor were employed. Powers Fasteners recommends a minimum safety factor of four but it appears that a higher safety factor has been used based on comparing the proof load of your field tests to the published ultimate tensile bond strength. This conservative design technique is not unusual. Adhesive anchoring problems typically reveal themselves immediately in a testing environment and do not manifest themselves over periods of time. For example, if the problem were dust in the hole, then the anchor would probably have moved during the proof loading procedure. If not properly mixed, the adhesive typically would not fully cure and remain somewhat tacky. This would also be apparent during the proof loading procedure.
- Overstress is a term more often associated with the applied torque to an adhesive anchor. The condition where the nuts have possibly been overtightened is described as an overstressed condition. In an ultimate adhesive anchor failure condition, the mode of failure can either be base material failure or more commonly, bond failure at the interface of the adhesive and the walls of the anchor hole. Anchor rod elongation is more likely to occur in deeper embedment depths (greater than 5") where the strength of the anchor rod material has effectively been developed.
- Since Powers Fasteners, Inc. is not the designer of this ceiling hanger plate assembly, an answer to this final question cannot be provided. Powers is not

aware of the different loads that may be acting on these anchors. In addition to dead loads, live loads such as wind, equipment and, personnel that potentially affect the applied load capacity should be considered.

While we believe that the existing anchors subjected to torques exceeding our recommended range may be acceptable, the only way to verify their suitability is by performing a proof load test. Successful proof load test results would indicate that the anchors are still functional. Nuts should then be reapplied to tested anchors with a torque that is within our recommended torque range.

Should you have any questions or require further information, please feel free to contact me at (914) 235-6300. Thank you for your continued interest in Powers Fasteners.

Very truly yours,
Powers Fasteners, Inc.

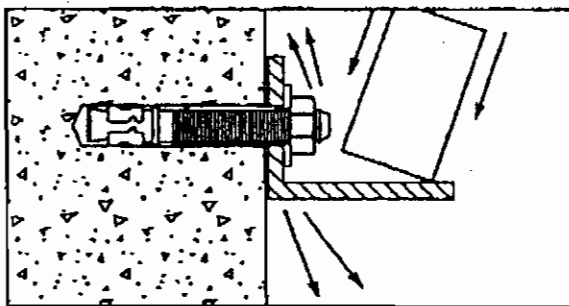


Ray Williamson
Engineering Manager

cc: William H. Alderman, P.E. – Powers Fasteners, Inc.
Stephen Rice – Powers Fasteners, Inc.
Jack Armour – Powers Fasteners, New England
Rich Tamulionis – Newman Renner Colony

Anchor Selection Guidelines

Shock loads are instantaneous, periodic loads of high intensity such as those applied by an automobile striking a guard rail support or a truck hitting a dock bumper.



Shock load

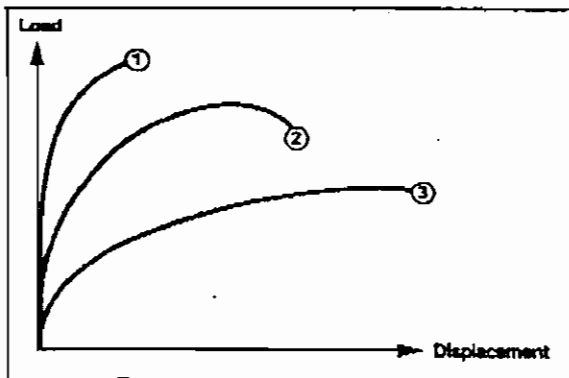
Standard industry practice with regard to safety factors varies depending upon the frequency and intensity of the load. However, safety factors for dynamic or shock load conditions may be as high as 10:1.

22.4 Anchor Behavior

The selection and specification of an anchor requires an understanding of basic anchor behavior or performance. A variety of performance attributes can be expected depending upon the type or style of anchor.

22.4.1 Displacement

As an anchor is loaded to its ultimate (failure) load capacity, displacement or movement of the anchor relative to the base material will occur. The amount of displacement will be effected by the anchor preload, the anchor material strength, the design of the expansion mechanism, and the strength of the base material. Typical load versus displacement curves are shown in the following diagram for three anchor types.



Curve 1 shows the typical performance of an adhesive type anchor. These anchors normally exhibit elastic behavior up to the ultimate load capacity. Performance will vary depending upon the type of adhesive used, the base material strength, and the strength of the anchor rod. A deformation controlled anchor such as a steel drop-in anchor may also exhibit this type of behavior although the ultimate load capacity will normally be

much less than that of an adhesive anchor. The compression force developed by a steel drop-in is usually very high when compared to a torque controlled anchor resulting in low displacement characteristics.

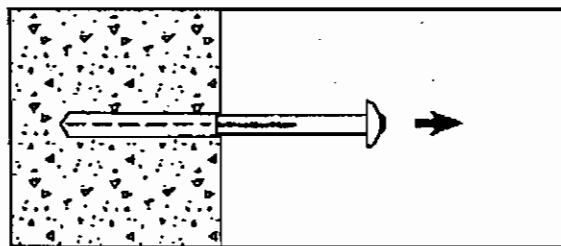
Typical performance of a torque controlled anchor such as the Power-Bolt or Power-Stud is shown in Curve 2. Displacement begins to occur after the initial preload in the anchor has been exceeded until the ultimate load capacity is achieved.

Anchors for use in light duty applications often exhibit the behavior shown in Curve 3. Once the working load has been exceeded, the anchor begins to displace or stretch until failure occurs.

22.4.2 Modes Of Failure

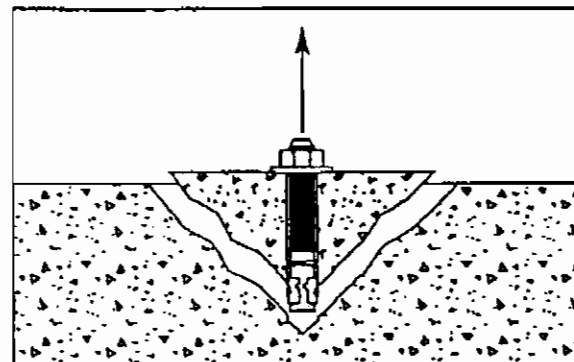
As an anchor is loaded to its ultimate capacity, the following modes of failure can occur.

22.4.2.1 Anchor Pullout



This type of failure occurs when the applied load is greater than the friction or compressive force developed between the anchor body and the base material. The anchor is unable to fully transfer the load to develop the strength of the base material. For adhesive anchors, this can occur with products which have a low bond strength.

22.4.2.2 Base Material Failure



When the applied load is greater than the strength of the base material, the material pulls out or fails. In concrete, a shear cone will be pulled, usually for anchors installed at a shallow embedment in the range of 4 to 5 anchor diameters of depth. The angle of the shear cone has been assumed to be 45°, however, this can vary up to 60° depending upon the anchor style and embed-